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Overactive bladder: treatment

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Treatment

The treatment of overactive bladder is patient-oriented and takes place in phases (figure 1). Initially conservative treatment, followed by more in-depth diagnostics if the response is insufficient and more invasive treatment. Treatment success is evaluated after 4 to 6 weeks. A bladder diary should be completed for 1 to 3 days before the consultation.

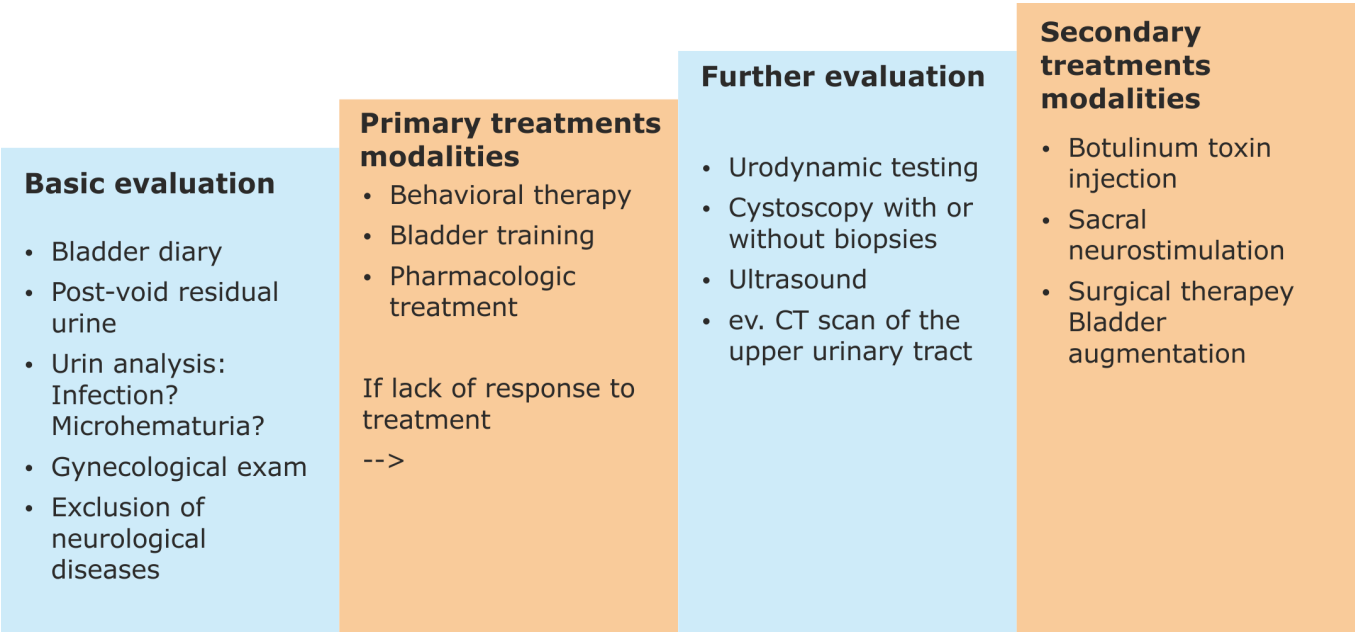

Figure 1: Phased approach to evaluation and treatment of overactive bladder.

The basic evaluation is aimed to diagnose and treat primary diseases that affect the bladder: colpitis, vulvitis, or urinary tract infection (cf. G02H), tumors in the lower pelvis (e.g., bladder papilloma, bladder cancer, large anterior uterine wall fibroids), urolithiasis, exclusion of neurological diseases (e.g. genital multiple sclerosis, dementia) or migration/exposure of foreign materials (mesh for treatment of prolapse or incontinence sling). All these pathologies can trigger an urge to urinate. In the case of micro-hematuria, the lower urinary tract and, in cases of no abnormal findings, the upper urinary tract should be investigated according to the guidelines.

Primary treatments modalities

The patient should learn:

- to delay micturition (behavioural therapy and bladder re-training)
- to drink sufficient quantities of liquids (drinking behavior)
- to avoid bladder-irritating substances (alcohol, substances containing caffeine and sweeteners, spicy food).

The patient should complete a bladder diary to record micturition volume, volume intake, frequency and urge and urge incontinence.

Cornerstones of treatment for overactive bladder

Bladder re-training and pelvic floor physiotherapy

With bladder re-training, the patient will be aware of the difference between the primary and secondary urge to urinate, and the bladder diary (cf. G02B) is analyzed by a specialist. With physiotherapy, reflexative pelvic floor activity is induced. The aid of transvaginal or peripheral electrostimulation lowers the threshold of the spontaneous detrusor overactivity (cf. G02C). A well-trained pelvic floor can also counteract detrusor muscle overactivity. [B07][B08][B09]

Pharmacotherapy

The aim is to reduce bladder tension and symptoms of urgency/ urge incontinence. This is achieved with anticholinergic agents, which have a spasmolytic or anticholinergic (parasympatholytic) effect, or β3 adrenergic agonists (table 1), which support the sympathetic storage function.

The effect of anticholinergics is the result of the antagonism on the M2 and M3 subtype of the muscarinic bladder receptors. Anticholinergics show a moderate improvement in quality of life versus placebo. Anticholinergic treatment should enable patients with overactive bladder to achieve a reduction of five micturitions and a reduction of four incontinence episodes per week. The drugs show a similar effect.

According to the Cochrane Database, anticholinergic drugs significantly improve the symptoms of overactive bladder versus bladder re-training alone (RR 0,73). A combination of anticholinergic medication and bladder re-training seems to be the ideal treatment option, at least with a wide confidence interval (RR 0,55; CI 0,32-0,93). Treatment is most often interrupted or discontinued due to side effects, most commonly due to dry mouth in approx. 30 % of patients treated. Extended-release products can be used to reduce this side effect. [P05]

Parasympatholytic side effects, including those arising outside the bladder, include dry mouth, constipation, visual disorders, central nervous system disorders, etc. In older patients in particular, it is important to watch out for a potential summation of anticholinergic side effects: confusion, disorientation, disturbed sleep and forgetfulness can arise! In cases of cognitive impairment or poly-pharmacy, particularly with co-medication with partial anticholinergic effects (e.g. antipsychotics, antidepressants, benzodiazepines, opiates, loop diuretics), anticholinergic agents should be used with the greatest caution.

Biotransformation by means of cytochrome P450 is an important component in the degradation of numerous substances, and the combination of several substances that inhibit cytochrome p450 can cause pronounced changes in plasma levels and intensify side effects.

Of all anticholinergic substances, only trospium chloride is not metabolized by cytochrome p450. Instead, it is excreted through the kidneys as an active metabolite. According to current data, trospium chloride and darifenacin do not lead to impaired cognitive function, although most studies have not been carried out in older patients.

Since 2014, a new β3 adrenergic agonist has been available on the Swiss market in the form of mirabegron (Betmiga®). The effect of β3 agonists is observed in the bladder wall, involving β3 adrenoreceptors. These receptors play a role in the bladder filling and storage phases.

Randomized studies have shown a significant decrease in incontinence episodes and increase in bladder capacity [P13].

The combination of anticholinergic and β3 adrenergic drugs is possible on the basis of current studies and leads to an additional improvement in the urge to urinate and incontinence [P07].

Hypertension, nasopharyngitis, itching, and headaches have been reported as side effects of β3 agonists. [P07][P13]

Mirabegron is contraindicated in cases of severe or insufficiently controlled hypertension. Mirabegron is a CYP2D6 inhibitor. Caution is advised with the additional administration of drugs that are metabolized via the same enzyme (e. g. metoprolol and desipramine). The effects of digoxin and warfarin are also enhanced during the administration of β3 agonists.

Table 1: Active substances, trade names and dosages of drug treatment for overactive bladder (as of January 2024)

Active substance	Products	Dosage, administration, treatment duration
Solifenacin	Vesicare®	5 mg initially once per day p.o.; can be increased to 10 mg per day
Tolterodine	Detrusitol®	2 mg or 4 mg once per day p.o.
Darifenacin	Emselex®	7,5 mg initially once per day p.o.; can be increased to 15 mg
Fesoterodine	Toviaz®	4 mg once per day p.o.; can be increased to 8 mg
Oxybutynin	Kentera® Vesoxx®	3,9 mg per day transdermal plaster; 2 plasters per week intravesical instillation of 10-30/40 mg per day
Trospium chloride	Spasmex® Spasmo-Urgenin®	2 x 20 mg per day p.o.; short half-life
Mirabegron (β3 agonist)	Betmiga®	25 mg initially once per day p.o.; can be increased to 50 mg per day

Injecting intravesical botulinum toxin into the detrusor muscle should only be considered after failure of conservative treatment. Bladder capacity can be increased through the “paralysis” of the bladder muscles. Urine retention is possible. However, the effect of intravesical botox decreases after 6 to 9 months and the injection can then be repeated. Tachyphylaxis has not been documented. The costs for intravesicular botulinum toxin injection for the treatment of overactive bladder have been covered by health insurance providers since January 1, 2015.

The effect of intravesical botox treatment is higher than that of drug-based treatment. In a study by Schurch et al., 2007, a subjective improvement was found in 88 % of patients, 86 % had no more incontinence and in 82 %, the urge to void had disappeared [P01].

Undesired side effects of intravesical botox treatment include temporary urine retention, increased residual urine volume (sometimes requiring intermittent self-catheterization) and urinary tract infection.

In certain cases, psychosomatic treatment may be indicated (the bladder as a “mirror of the soul”).

If genital atrophy is found, local estrogenization in the form of cream, pessaries or vaginal tablets can improve symptoms. [P01]

The following products are available (as of January 2024): Estriol e.g. Ovestin® Oestrogynaedron® and Blissel®, as ultra-low-dose estriol preparations 0,03 mg Kadefemin® or Gynoflor®; estradiol e.g. Vagifem® and Estring®; promestriene e.g. Colpotrophine® (available from Zurich Cantonal Pharmacy). Local hormones are inserted vaginally twice a week in the evening. At this dosage, progesteron supplementation to avoid endometrial hyperplasia is not required.

In patients with a history of breast cancer or other hormone-dependent tumors, caution is recommended and muco-adhesive gels without hormone supplementation are the first choice in the treatment of urogenital atrophy and bladder irritation. If local estrogenization is unavoidable, the lowest dosed preparations should be used.

The role of estrogens in regard to overactive bladder has not been fully clarified yet. Some prospective studies have shown a benefit as a result of local estrogenization. Systemic estrogen therapy has shown no improvement to urge incontinence [B06].

Flavoxate (Urispas®) is an older spasmolytic drug and moderate calcium channel blocker that inhibits phosphodiesterase and has a local anesthetic effect. Its effect on symptoms of urgency is documented inconsistently, and side effects are low, emphasizing its role as an additive. [B06]

Antidiuretic hormones (desmopressin) stimulate water reabsorption in the renal medulla and can be indicated for nocturnal enuresis or nocturia in the context of overactive bladder, with no other causes (cardiac, endocrinological, or drinking habits). In older patients, it is important to monitor water and electrolyte balance and blood pressure. There is currently no longer a preparation authorised for this purpose in Switzerland.

If the above listed medicamentous treatment options fail, neuro-stimulation has proven to be an effective treatment modality. A continuous electrical impulse, which is transferred through surface or needle electrodes on the lower leg to the tibial nerve or from a pacemaker implanted in the buttocks and via fine wires into the sacral root of S2/S3, is used to achieve a persistent reduction in the imperative urge to void, a reduction of the micturition frequency and urge incontinence in 50 to 70 % of patients – namly in patients that showed responses to test electrodes during a 1 to 2 week test phase [P06].

Bladder augmentation or neo-bladder: As a last resort, in cases of chronic changes to the bladder wall, for example such as radiogenic contracted bladder, the bladder capacity can be increased using an ileum or colon loop transplant.

If the Credé maneuver does not lead to sufficient voiding of the bladder, the neo-bladder must be voided by means of a life-long self-catheterization. [P06]

Practice and research

The etiology of overactive bladder is still not fully understood yet; there are myogenic, neurogenic (peripheral and central) and chronic infectious theories. In the future, the role of the microbiome may provide clarification with regard to the pathogenesis. There are also hints that OAB and bladder pain syndrome may have similarities in regard to the local immune responses. [P14]

Simplified cell-to-cell communication, for example via increased connexon formation, as with the myocardium, is presumed but has not yet been proven for OAB. Bladder pain syndrome (BPS; previously: interstitial cystitis or painful bladder syndrome), where the symptoms of overactive bladder are coupled with a reduced bladder capacity and painful micturition as well as a normal urine status, is attracting increased attention (criteria: see www.ESSIC.org). BPS is found with endometriosis in 30 % of cases. The quality of life of the women with BPS is affected to a higher degree than the one that suffer from overactive bladder alone.

ESSIC International Society for the Study of Bladder Pain Syndrome (BPS)

Further substances are being investigated for the treatment of overactive bladder, in particular where painful micturition co-exists: substances affecting the central nervous system such as gabapentin (e.g., Neurontin®), tramadol (Tramal®), and NK1 receptor antagonists (e.g., Aprepitant®) belong to these newer substances, on which little clinical data has been published up to date.
Pentosan polysulphate (Elmiron®) has recently been approved in Switzerland for bladder pain syndrome as a supportive medication to build up the urothelial protective layer.

Literature

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B01]	The Standardisation of Terminology of Lower Urinary Tract Function	Abrams P et al	Neurourology and Urodynamics	21	167-178	200
[B02]	Textbook of female Urology and Urogynaecology	Cardozo/ Staskin	Informa Verlag	Second Edition, Volume I und II		
[B03]	Urogynecology and reconstructive pelvic surgery	Walters Mark D, Karram Mickey M		Fourth Edition		201
[B04]	Overactive Bladder Syndrome and Nocturia	South Mary, Karram Meckey M			513-541	201
[B05]	NICE clinical guidelines	National Institute for Health and Clinicial Excellence				201

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B06]	Oestrogens and overactive bladder	Robinson D, Cardozo L, Milsom I, Pons ME, Kirby M, Koelbl H, Vierhout M	Neurourol Urodyn	33(7)	1086-1091	201
[B07]	Efficacy of pelvic floor muscle training in women with overactive bladder syndrome: a systematic review	Monteiro S, Riccetto C, Afaujo A, Galo L, Brito N, Botelho S	Int Urogynecol J			201
[B08]	The effecto fo EMG biofeedback assisted pelvic floor muscle therapy on symptoms of the overactive bladder syndrome in women: a randomized controlled trial	Voorham CJ, Wachter SD, Van de Bos TWL, Putter H, Nijeholt GAL, Voorham PJ	Neurourol Urodyn	36(7)	1796-1803	201
[B09]	Electrical stimulation with non-implanted electrodes for overactive bladder in adults	Stewart F, Gameiro LF, El Dib R, Gameiro MO, Kapoor A, Amaro JL	Cochrane Database Syst Rev	12		201

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[P01]	Botulinum toxin A improves the quality of life of patients with neurogenic urinary incontinence	Schurch B, Denys P, Kozma CM	Eur Urol	52(3)	850	2007
[P02]	Botulinum toxin injections for adults with overactive bladder syndrome	Duthie JB, Vincent M, Herbison GP, Wilson DI, Wilson D	Cochrane Database Syst Rev	7	12	2011
[P03]	Systematic review and meta-analysis: do clinical trials testing antimuscarinic agents for overactive bladder adequately measure central nervous system adverse events?	Paquette A, Gou P, Tannenbaum C	J Am Geriatr Soc	59(7)	1332-1339	2011

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[P04]	Efficacy and tolerability of mirabegron, a β(3)-adrenoceptor agonist, in patients with overactive bladder: results from a randomised European-Australian phase 3 trial	Khullar V, Amarenco G, Angula JC, et al	Eur Urol	63	283	2013
[P05]	Anticholinergic drugs versus non-drug active therapies for non-neurogenic overactive bladder syndrome in adults	Rai BP, Cody JD, Alhasso A, Stewart L	Cochrane Database Syst Rev	12	12	2012
[P06]	Long-term durability of percutaneous tibial nerve stimulation for the treatment of overactive bladder	MacDiarmid SA, Peters KM, Shobeiri SA, et al.	J Urol	183	234	2010

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[P07]	Efficacy and Safety of Mirabegron Add-on Therapy to Solifenacin in Incontinent Overactive Bladder Patients with an Inadequate Response to Initial 4-Week Solifenacin Monotherapy: A Randomised Double-blind Multicentre Phase 3B Study (BESIDE)	Drake MJ, Chapple C, Esen AA, et al	Eur Urol	70	136	2016
[P08]	ombination treatment with mirabegron and solifenacin in patients with overactive bladder: exploratory responder analyses of efficacy and evaluation of patient-reported outcomes from a randomized, double-blind, factorial, dose-ranging, Phase II study (SYMPHONY)	Abrams P, Kelleher C, Staskin D, et al	World J Urol			2016

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[P09]	Tolerability and persistence in a large, prospective case series of women prescribed mirabegron	Duckett J, Balachandran A	Int Urogynecol J	27	1163	2016
[P10]	A systematic review of the efficacy and safety of desmopressin for nocturia in adults	Ebell MH, Radke T, Gardner J	J Urol	192(3)	829-835	2014
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[P13]	Use of mirabegron in treating overactive bladder	Bhide AA, Digesu GA, Fernando R, et al	IUJ	23	1345-1348	2012
[P14]	Local immune response in bladder pain syndrome/ interstitial cystitis ESSIC type 3C	Gamper M, Viereck V, Eberhard, J, Binderj, Moll C, Welter J, Moser R	Int Urogynecol J	24(12)	2049-2057	2013